ChildLens: An Egocentric Video Dataset for Activity Analysis in Children

Nele-Pauline Suffo[1], Pierre-Etienne Martin[2], Anas Suffo[1], Daniel Haun[2], & Manuel Bohn[1, 2]

[1] Institute of Psychology in Education, Leuphana University Lüneburg, Universitätsallee 1, 21335 Lüneburg, Germany

[2] Comparative Cultural Psychology, Max Planck Institute for Evolutionary Anthropology, Deutscher Platz 6, Leipzig, 04103, Germany

Author Note

Correspondence concerning this article should be addressed to Nele-Pauline Suffo, Universitätsallee 1, 21335 Lüneburg, Germany. E-mail: nele.suffo@leuphana.de. Phone: +49 4131 677-1905.

Abstract

We present ChildLens, an egocentric video and audio dataset that captures naturalistic everyday experiences in children aged 3–5 years, including detailed activity labels. A total of 109 hours of experiences were recorded from 62 children in their home environment using a 140° wide-lens camera equipped with a microphone integrated in a child-friendly vest. Annotations include five location classes and 14 activity classes, covering audio-only, video-only, and multimodal activities. Captured through the camera on chest height, ChildLens provides a rich resource for analyzing children's daily interactions and behaviors. We provide an overview of the dataset, the collection process, and the labeling strategy. Finally, we present benchmark performance of two state-of-the-art models on the dataset: the Boundary-Matching Network for temporal activity localization and the Voice Type Classifier for detecting and classifying speech in audio. The ChildLens dataset will be freely available for research purposes via an institutional repository (listed on the ChildLens website). It provides rich data to advance computer vision and audio analysis techniques and thereby removes a critical obstacle to studying the context of child development.

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Introduction

In developmental psychology, everyday experiences play a central role when theorizing about the causes and dynamics of developmental change (Carpendale & Lewis, 2020; Heyes, 2018; Piaget, 1964; Rogoff, Dahl, & Callanan, 2018; Smith, Jayaraman, Clerkin, & Yu, 2018; Tomasello, 2009; Vygotsky, 1978). Famously, the two central processes in Piaget's theory of cognitive development - assimilation and accommodation - describe how children's cognitive abilities change in line with the experiences they make (Piaget, 1964). Vygotsky emphasized the role of social interactions between children and (knowledgeable) adults for the acquisition of culturally relevant knowledge (Vygotsky, 1978). Contemporary theorists rest on similar ideas. For example, Tomasello (2009) pointed out how everyday social interactions, particularly those involving shared intentionality, foster uniquely human forms of communication, cooperation and cognition. For Heyes (2018), culturally evolved "cognitive gadgets" are transmitted via language in conversations between adults and children.

These broad theoretical accounts are based on empirical support. Reviewing evidence for all these theoretical accounts is beyond the scope of this paper, so we want to point out only a few examples from the domain of language development. Across languages and cultural settings, the amount of language children hear is related to their language development (Bergelson et al., 2023). There is also a relationship between the amount of conversational turn-taking children are engaged in and their vocabulary growth (Donnelly & Kidd, 2021; Ferjan Ramírez, Lytle, & Kuhl, 2020). Roy, Frank, DeCamp, Miller, and Roy (2015) showed that words that are heard in distinct contexts at distinct times are more likely to be learned. Ruffman et al. (2023) used head-mounted video cameras to study how repeated behaviors in everyday life correlate with the acquisition of mental state vocabulary. Taken together, such studies illustrate how important it is to study naturalistic everyday experiences to understand children's development. However, studies linking everyday experience and development are still vastly underrepresented (De Barbaro & Fausey, 2022; Rogoff et al., 2018) as they come with a set of unique challenges.

Perhaps the most significant obstacle in this field is the extensive amount of data needed to comprehensively study children's everyday experiences. Traditional methods, such as manual annotation, are time-consuming and impractical for large-scale datasets. To address this, computer vision or Natural Language Processing (NLP) models offer scalable solutions for analyzing social interactions and behaviors. For instance, OpenPose (Cao, Hidalgo, Simon, Wei, & Sheikh, 2018) allows the tracking of human body, face, and hand poses which provides valuable insights into gestures and engagement. YOLOv8 (Redmon, Divvala, Girshick, & Farhadi, 2015) offers efficient object detection and models like I3D (Carreira & Zisserman, 2017) provide an automated solution for classifying activities in video data. For audio, Wave2Vec 2.0 (Baevski, Zhou, Mohamed, & Auli, 2020) provides robust speech-to-text and speech representation capabilities, enabling the study of conversational dynamics. Together, these models facilitate the efficient analysis of multimodal data. However, even the best model architecture needs diverse, high-quality datasets to learn from. A notable example of such a dataset is ImageNet (Russakovsky et al., 2014); in fact, the release of this dataset sparked the development of some of the most prominent computer vision models available today. Similarly, expanding publicly available datasets in developmental psychology could accelerate progress in studying children's everyday experiences.

Several publicly available datasets have made valuable contributions to our understanding of children's social and communicative behavior. For example, the SAYCam dataset (Sullivan, Mei, Perfors, Wojcik, & Frank, 2021) provides audio-video recordings from 3 infants (6–32 months) who wore head-mounted cameras over two years, to capture naturalistic speech and behaviors. Similarly, the DAMI-P2C dataset (Chen, Alghowinem, Jang, Breazeal, & Park, 2023) includes audio and video recordings of parent-child interactions during story reading, with annotations for body movements in a controlled environment. The MMDB dataset (Rehg et al., 2013) offers multimodal data (audio, video, physiological) of children (15–30 months) engaged in semi-structured play interactions, recorded in a lab. Another example is the UpStory dataset (Fraile et al., 2024), which features audio and video of primary school children (8-10 years) in dyadic storytelling interactions, also recorded in a lab setting. Additionally, the BabyView dataset (Long et al., 2024) provides high-resolution, egocentric video of children aged 6 months to 3 years, recorded at home and in preschool environments, with annotations for speech transcription and pose estimation. Whereas these datasets vary in age, setting, and target behaviors, they collectively highlight the need for more naturalistic, at-home datasets that can capture the full range of children's daily activities.

To address this gap, we introduce the publicly available ChildLens dataset, which focuses on activity annotations for children aged 3–5 years. The dataset consists of 108.58 hours of video and audio recordings collected from 62 children in their home environment through a 140° wide-lens camera, integrated in a child-friendly vest (see also Zahra, Martin, Bohn, and Haun (2024)). It includes detailed annotations for five location classes and 14 activity classes, which are further categorized based on whether the child is interacting alone or with others. Designed to support research in developmental psychology and computer vision, the ChildLens dataset offers a rich resource for advancing multimodal learning and studying the full spectrum of children's daily activities. As of now, 51% (around 55 hours) of the dataset are annotated. The annotation process is ongoing.

Dataset Generation

This section outlines the steps taken to create the ChildLens dataset. We provide detailed information on the video collection process, the labeling strategy employed, and the generation of activity labels.

Step 1: Collection of Egocentric Videos

The ChildLens dataset consists of egocentric videos recorded by children aged 3 to 5 years over a period of 12 months. A total of 62 children from families living in a mid-sized city in Germany, participated in the study. The videos were captured at home using a camera embedded in a vest worn by the children, as shown in figure 1. This setup allowed the children to move freely throughout their homes while recording their activities. The camera, a *PatrolEyes WiFi HD Infrared Police Body Camera*, was equipped with a 140° wide-angle lens and captured the space in front of the child with a resolution of 1920x1080p at 30 frames per second. The camera also recorded high-quality audio, allowing us to capture the child's speech and other sounds in the environment.

In order to obtain a decent coverage for the different activity classes we planned to annotate, we handed parents a short checklist of activities to record. The focus was on capturing everyday activities that children typically engage in. Parents were therefore asked to include the following elements in the recordings:

- Child spends time in different rooms and performs various activities in each room
- Child is invited to read a book together with an adult
- Child is invited to play with toys alone
- Child is invited to play with toys with someone else (adult or child)
- Child is invited to draw/craft something

Step 2: Creation of Labeling Strategy

To create a comprehensive labeling strategy for the ChildLens dataset, we first defined a list of activities that children typically engage in. This list was inspired by previous research on activities that children are known to participate in (Ginsburg, Committee on Communications, & Committee on Psychosocial Aspects of Child and

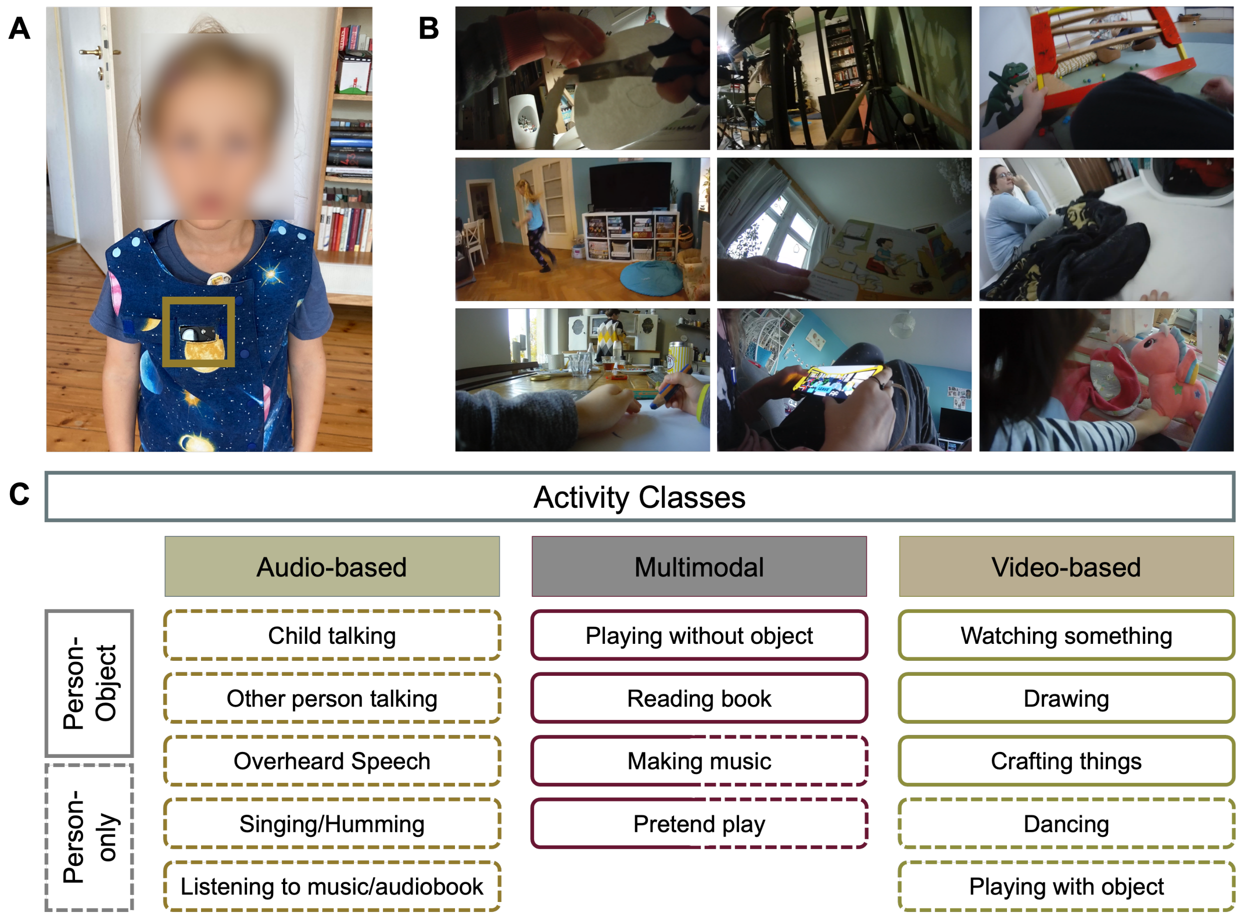

Figure 1. **A** – Vest with the integrated camera worn by the children, **B** – Collage of 9 screenshots showing activities (top left to bottom right): crafting, dancing, drawing, making music, reading a book, watching something, playing with an object, playing without an object, and pretend play, **C** – Activity classes in the ChildLens dataset.

Family Health, 2007; Hofferth & Sandberg, 2001). From this, we derived a detailed catalog of activities that were likely to be captured in the videos and chose to make the activity classes more granular by distinguishing between activities like "making music" and "singing/humming" or "drawing" and "crafting things".

After an initial review of the videos, we decided to add another class "overheard speech" to capture situations in which the child is not directly involved in a conversation but can hear it. We also added "pretend play" as a separate class to capture situations in which the child is engaged in imaginative play. This approach allowed us to capture the diversity of activities that children engage in and create a comprehensive dataset for activity analysis.

Step 3: Manual Labeling Process

Before the actual annotation process, a setup meeting was held to introduce the annotators to the labeling strategy. To familiarize themselves with the task, the annotators were assigned 25 sample videos to practice and gain hands-on experience. These initial annotations were reviewed by the research team, and feedback was provided to refine the approach. A total of three feedback loops were conducted to ensure that the annotators follow the labeling strategy properly.

The videos were manually annotated by native German speakers who watched each video and labeled the activities present in the footage. Annotators marked the start and end points of each activity. For audio annotations, we implemented a 2-second rule for all audio-based activity classes: if the break between two instances was two seconds or less, it was considered a single event; breaks longer than two seconds split the activity into separate instances.

Dataset Overview

Activity Classes. The ChildLens dataset includes 14 activity classes and 5 location classes. A brief description of each class can be found in table 1. The location classes describe the current location of the child in the video and include *livingroom*, *playroom*, *bathroom*, *hallway*, and *other*. The activity classes are categorized based on the child's interactions within the video and can be divided into *person-only* activities (e.g. "child talking", "other person talking"), and *person-object* activities (e.g. "drawing", "playing with object"). These activities are further categorized into *audio-based*, *visual-based*, and *multimodal* activities, as presented in figure 1. Below is an overview of the different activity types:

- **Audio-based activities**: *child talking*, *other person talking*, *overheard speech*, *singing / humming*, *listening to music / audiobook*
- **Visual-based activities**: *watching something*, *drawing*, *crafting things*, *dancing*, *playing with object*
- **Multimodal activities**: *playing without object*, *reading book* , *making music*, *pretend play*

Statistics. The ChildLens dataset comprises of 354 video files with a total of 108.58 hours recorded by 62 children aged 3 to 5 years (M=4.56, SD=0.93), including 32 female and 30 male children. It contains 30.02 hours from children aged 3, 41.91 hours from children aged 4, and 36.66 hours from children aged 5. The video duration per child varies between 4 and 303 minutes (M=122.11, SD=56.63). A detailed distribution of the video duration per child is shown in figure 2.

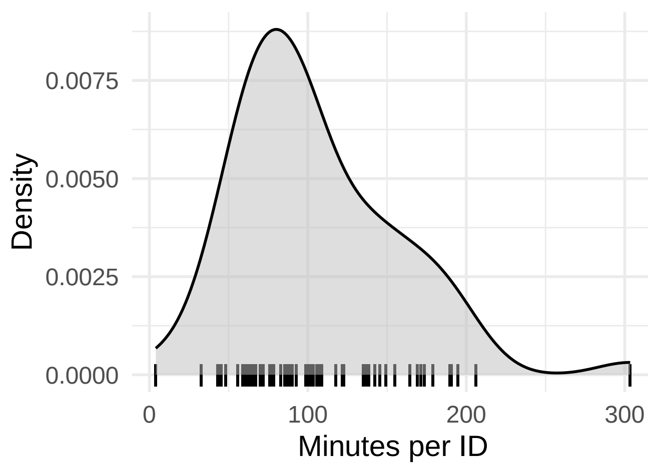

Figure 2. Distribution of video recording duration (in minutes) per child ID.

This diverse dataset includes a varying number of instances with different total duration across the 14 activity classes. While annotations are still ongoing, the current

Table 1

Activity classes in the ChildLens dataset (first 12 child-centered, last 2 other-person-centered).

Activity Class	Description
Child talking	Talking to themselves or to someone else.
Singing/Humming	Singing or humming a song or a melody.
Listening to music/audiobook	Listening to music or an audiobook.
Watching something	Watching a movie or video on either a screen or a device.
Drawing	Drawing or coloring a picture.
Crafting things	Engaged in a craft activity, such as making a bracelet.
Dancing	Dancing to music or moving to a rhythm.
Playing with object	Playing or interacting with an object, such as a toy or a ball.
Playing without object	Playing without an object, such as playing hide and seek
Pretend play	Engaged in imaginative play, such as pretending to be a doctor.
Reading a book	Reading a book or looking at pictures in a book.
Making music	Playing a musical instrument or making music in another way.
Other person talking	Another person is talking to the child.
Overheard Speech	Conversations the child can hear but is not directly involved in.

annotations (51% annotated) include between 2.47 and 938.04 hours per class. Depending on the activity class, the total duration for each activity class can consist of many shorter instances or fewer longer instances.. For example, audio-based activities like “child talking” may last only a few seconds, while activities like “reading a book” can span several minutes. The total number of instances and summed duration for all activity classes is available in table 4 in the appendix.

Data Availability. The ChildLens dataset will be made available to scientists for research purposes. It includes video and audio recordings, along with activity labels. Due to the sensitive nature of the data—recordings of children in their homes—access will be restricted.

Researchers can submit requests for access, which will be carefully reviewed to ensure proper handling and compliance with privacy standards. Please contact ra@eva.mpg.de to request access to the dataset. As noted above, the annotation process is still ongoing and the dataset will be updated regularly. A brief project overview, along with the latest dataset version can be found on the ChildLens website.

Exhaustive multi-label annotations. The dataset provides detailed annotations for each video file. These annotations specify the child’s current location within the video, the start and end times of each activity, the activity class, and whether the child is engaged alone or with somebody else. For every person involved in the activity, we capture age class and gender. If multiple activities occur simultaneously in a video, each activity is labeled individually. For example, if a segment shows a child “reading a book” while also “talking,” two separate annotations are created: one for “reading a book” and another for “child talking.” This exhaustive labeling strategy ensures that each activity is accurately represented in the dataset.

Benchmark Performance

In this chapter, we present the results of applying two model architectures to the currently annotated subset of the ChildLens dataset for two specific tasks: temporal activity localization using video data and voice type classification using audio data. For temporal activity localization, we used the Boundary-Matching Network (BMN) model (Lin, Liu, Li, Ding, & Wen, 2019), a state-of-the-art approach in this domain, which we trained from scratch on the unique activity classes in the ChildLens video data. For voice type classification, we applied and trained the Voice Type Classifier (VTC) (Lavechin, Bousbib, Bredin, Dupoux, & Cristia, 2020), also state-of-the-art, using three different evaluation setups. Both models provide initial results and establish a benchmark for future research. The model architectures for both, BMN and VTC, are displayed in table 4 and table 5 in the appendix, respectively.

All calculations were conducted on a Linux server equipped with two NVIDIA Quadro RTX 8000 GPUs, each with 48 GB of VRAM (CUDA version 12.5, driver version 555.42.06). The server had 48 CPU cores and 187 GB of RAM. The experiments were implemented in Python 3.8.18 for Temporal Activity Localization and Python 3.8.19 for Voice Type Classification using PyTorch (version .4.1+cu121 12.1), MMCV (version 2.2.0), and MMAction2 (version 1.2.0).

Temporal Activity Localization

The Boundary-Matchin Network generates action proposals by assigning confidence scores for predicted activity start and end boundaries. The architecture consists of two main components: (1) a proposal generation network, which identifies candidate proposals, and (2) a proposal evaluation network, which provides confidence scores for these proposals. The model prioritizes proposals with high recall and high temporal overlap with ground truth. BMN performance is evaluated using Average Recall (AR) and Area Under the Curve (AUC) metrics. AR is computed at various Intersection over Union (IoU) thresholds and for different Average Numbers of Proposals (AN) as AR@AN, where AN ranges from 0 to 100. AR@100 reflects recall performance with 100 proposals per video, while AUC quantifies the trade-off between recall and number of generated proposal. On the ActivityNet-1.3 test set (Heilbron, Escorcia, Ghanem, & Niebles, 2015), BMN demonstrates effective activity localization with an AR@100 of 72.46 and an AUC of 64.47. A subsequent reimplementation of the model achieved slightly improved results (Xin, 2020), as displayed in table 2, further confirming its robustness.

Data Preparation. The BMN implementation, including video preprocessing and model training, was conducted using the MMAction2 toolbox (Contributors, 2020). Data preparation involved several key steps, such as raw frame extraction and the generation of both RGB and optical flow features for each video. Before training the model, we analyzed the distribution of activity instances across the classes for the annotated videos to assess the data's sufficiency for both training and testing (see table 4 in the appendix).

Our analysis highlighted a significant class imbalance in the dataset, both in terms of instance count and the total duration of recordings. Given the primary goal of establishing initial benchmark results, no data augmentation methods were employed to mitigate this imbalance. Instead, we focused on the more frequent activity classes, which also had the longest duration: "Playing with Object" (30.82 hours of recording), "Reading a Book" (7.45 hours of recording) and "Drawing" (6.1 hours of recording).

For feature extraction and model training optimization, the videos were divided into clips of 4000 frames (each clip corresponds to approx. 2 minutes and 13 seconds). We then split these annotated clips into training, validation, and test subsets, using an 80-10-10 split. The training set was used for model optimization, the validation set guided hyperparameter tuning and overfitting prevention, and the test set was reserved for evaluating the model's generalization ability on unseen data.

Table 2

BMN performance on ActivityNet-1.3 and ChildLens datasets, showing Average Recall (AR) at different numbers of generated proposals (AR@1, AR@5, AR@10, AR@100) and Area Under the Curve (AUC).

Dataset	AR@1	AR@5	AR@10	AR@100	AUC
ActivityNet-1.3	33.6	49.9	57.1	75.5	67.7
ChildLens	**50.6**	**56.7**	**64.2**	**77.5**	**71.4**

Note. Values are reported as percentages.

Implementation Details. The BMN model was trained from scratch on the ChildLens dataset to predict the start and end boundaries of activity classes in the videos. The model was implemented using MMAction2, an open-source toolbox for video understanding based on PyTorch (Contributors, 2020). The Adam optimizer was used with a learning rate of 0.001 and a batch size of 4. To avoid overfitting, early stopping based on validation loss was applied during training.

Evaluation. The performance of the BMN model on the ChildLens dataset, compared to its evaluation on ActivityNet-1.3, is summarized in table 2, with Average Recall (AR) reported for different number of generated proposals and Area Under the Curve (AUC) for both datasets. Across all proposal settings (AR@1 to AR@100), the BMN model achieves higher performance on ChildLens compared to ActivityNet. This result is promising given that ChildLens has significantly fewer activity classes than ActivityNet (n=200), and its videos are also shorter on average. In this context, evaluating at a lower number of proposals is more meaningful. Even when limited to just 10 proposals per video, the model achieves a strong AR@10 of 64.23 on ChildLens, underscoring both the quality and consistency of our annotations. These benchmark results suggest that our annotations provide a solid basis for training and evaluating temporal action proposal models.

Voice Type Classification

Voice Type Classification is the task of identifying audio utterances and assigning them to predefined classes. In line with previous work, we focus on the following classes: Key Child (`KCHI`), Other Child (`CHI`), Male Speech (`MAL`), Female Speech (`FEM`), and Speech (`SPEECH`), as defined by the Voice Type Classifiers (VTC) output classes (Lavechin et al., 2020). Its architecture is composed of a SincNet layer, two bi-directional LSTMs, and three feed-forward layers. The model takes a 2-second audio chunk as input and outputs a score between 0 and 1 for each class. The VTC was originally trained on the BabyTrain dataset which includes 260 hours of child-centered audio in multiple languages from children mostly aged 0-3 years.

The model architecture utilizes the open-source pyannote library for speaker diarization (Bredin, 2023; Plaquet & Bredin, 2023) which provides pre-trained models and pipelines for various audio processing tasks. By adjusting the model architecture and training process using pyannote, we evaluated the ChildLens dataset's quality through three distinct VTC training setups: In the first setup, we applied the pretrained VTC model directly to the ChildLens dataset. In the second setup, we finetuned the VTC model on the ChildLens data. In the last setup, we trained the VTC model from scratch using only the ChildLens dataset. These setups test the dataset's annotation consistency and standalone value, with performance measured by the F_1-measure, a VTC-specific parameter (Bredin, 2017), which combines precision and recall.

Data Preparation. We used the following mapping strategy to align our audio-based activity classes with the VTC's output classes. Minutes in parentheses indicates the total duration of annotated audio for each class.

- Child talking & Singing/Humming → **`Key Child`** (963.44 min)

- Other person talking:
  - If age = `"Child"` → **`Other Child`** (53.55 min)
  - If age = `"Adult"` & gender = `"Female"` → **`Female Speech`** (492.96 min)
  - If age = `"Adult"` & gender = `"Male"` → **`Male Speech`** (223.41 min)
- Overheard Speech, Child talking, Singing/Humming, Other person talking → **`Speech`** (2158.24 min)

"Listening to music/audiobook" was excluded, as it's often background audio, lacks speaker age/gender details, and includes music irrelevant to VTC. "Overheard Speech" refers to speech not directed at the key child and was mapped to SPEECH because no age/gender information were collected in the annotation process. This may underestimate VTC performance, for cases when the VTC correctly predicts a woman speaking as FEM, but is labeled as SPEECH. We retain this mapping, as re-annotation would be time-consuming. Moreover, the class is conceptually incompatible with the VTC output classes: it captures the recipient of speech rather than the speaker type. It is intended for more advanced models (for instance NLP architectures) that can infer not just who is talking, but also to whom, based on semantic content.

Implementation Details. For the first and second setup, using the pretrained and finetuned VTC models, we used the original model architecture. For the third setup, VTC trained from scratch VTC_{cl}, we made slight adjustments to the model architecture to boost performance. We increased the RNN hidden size from 128 to 256, the number of RNN layers from 3 to 4, added dropout (0.1), extended the input duration from 2.0 to 3.0 seconds and doubled the number of steps per epoch from 1 to 2. All other parameters remained unchanged.

Evaluation. Table 3 shows F_1-scores for three VTC setups on the ChildLens dataset: the original model VTC_{og}, the finetuned model VTC_{ft}, and the model trained from scratch VTC_{cl}, compared to the benchmark dataset. VTC_{og} scores 45.70, slightly

Table 3

Comparison of Voice Type Classifier (VTC) performance on the ACLEW-Random dataset and the three setups utilizing the ChildLens dataset. The evaluation metrics are reported for the original VTC model (VTC-OG), the VTC finetuned on ChildLens (VTC-FT) and the VTC trained from scratch on ChildLens (VTC-CL). The table reports the F1 score per class and the average F1 score (AVG).

Dataset	Model	KCHI	CHI	MAL	FEM	SPEECH	AVG
ACLEW-Random	VTC-OG	NA	NA	NA	NA	NA	NA
ChildLens	VTC-OG	NA	NA	NA	NA	NA	NA
7.63 49.75 40.97 84.26 51.24		NA	NA	NA	NA	NA	NA
ChildLens	VTC-FT	NA	NA	NA	NA	NA	NA
ChildLens	VTC-CL	NA	NA	NA	NA	NA	NA

below the benchmark, with highest performance on `KCHI` (68.70) and lowest on `CHI` (8.20). Both VTC_{ft} and VTC_{cl} achieve substantially higher scores (55.20 and 50.60, respectively), indicating significant improvement compared to the VTC_{og} ChildLens performance. These results provide strong evidence for the high quality of the ChildLens annotations and their suitability for training and evaluating models with different architectures.

The low `CHI` scores are primarily due to frequent misclassifications as `KCHI`. This likely results from the vocal similarity between the key child and other children in the ChildLens dataset. In many cases, distinguishing between them may rely more on acoustic cues like proximity to the microphone than on vocal characteristics. In contrast, the original VTC dataset includes audio from infants and toddlers, where the distinction between the key child (often babbling or younger) and other speakers may be more pronounced, making classification tasks easier. Importantly, these scores are expected to improve as more annotated data become available; as mentioned earlier, the annotation process is still ongoing. Figure 3 visualizes how model predictions compare to the ground truth annotations for the VTC_{cl} model.

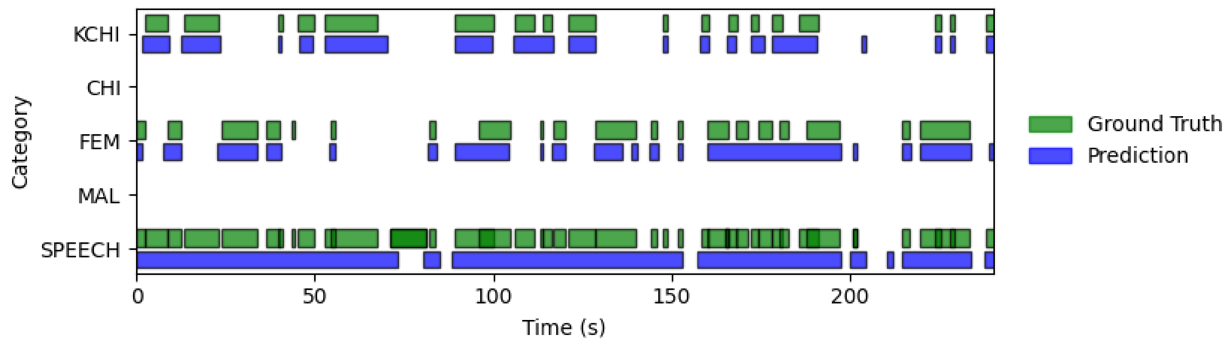

Figure 3. VTC_{cl} Predictions compared to Ground Truth Annotations

General Discussion

We present the ChildLens dataset, a unique egocentric video-audio dataset that documents naturalistic everyday experiences in preschool children. This dataset is particularly distinctive due to its diversity in terms of the number of children it includes and the variety of activity labels it covers. By encorporating both visual and auditory data, the ChildLens dataset provides comprehensive annotations for a broad spectrum of key activities, including multi-modal social interactions. These annotations support the training and evaluation of models for the automatic analysis of children's activities and therefore allow for scaling up data collection. A rich and representative dataset like ChildLens is essential for understanding individual differences in children's daily experiences and how they relate to different cognitive and social outcomes.

In comparison to other freely available datasets, the ChildLens dataset stands out due to its broad age span and diverse set of activity labels. Most existing datasets either focus on toddlers, are limited to dyadic interactions or were recorded in lab settings, with all of them lacking a comprehensive range of activity labels. Furthermore, most of these datasets either capture only audio or video. In contrast, ChildLens includes naturalistic recordings from children's home environments, collected over an extended period, and features a wide variety of activity types. Particularly noteworthy is the activity class `Overheard Speech` which captures speech that children can hear but are not directly involved in. This class is important for studying the impact of overheard speech on children's cognitive development - an area that has largely relied on labor-intensive manual annotations due to the difficulty of automatically distinguishing between child-available and child-directed speech (Bergelson et al., 2023). Existing models, like the classifier developed by Bang, Kachergis, Weisleder, and Marchman (2023) could be enhanced by incorporating visual features - such as eye contact or the use of gestures - in addition to audio inputs. The ChildLens dataset also captures whether children are engaged in activities alone or with others and provides basic demographic information - age, sex -about all individuals involved.

The usefulness of the ChildLens dataset is demonstrated by its successful application to well-established models. For example, the Voice Type Classifier for audio classification achieves performance comparable to the previous benchmark dataset, while the Boundary-Matching Network produces robust results for activity localization, consistent with its performance on widely used datasets such as ActiviyNet. One way of using the ChildLens dataset to advance methodological development is through multi-method approaches. For example, activity localization could be further enhanced by incorporating object identification, allowing for better tracking of the objects children interact with during daily routines. Such an approach has been used in adult-focused studies (Kazakos, Huh, Nagrani, Zisserman, & Damen, 2021). Research by Bambach, Lee, Crandall, and Yu (2015) also emphasizes the importance of hand detection in egocentric video for activity recognition. Their use of Convolutional Neural Networks (CNNs) for hand segmentation demonstrates how such techniques can help differentiate between activities. To apply a similar approach to the ChildLens dataset, we would first need to run a pretrained hand detection model on the dataset. Afterwards, we would train a CNN to classify frames containing hands into activity classes, following the method described by Bambach et al.

(2015). If the performance of the hand detection performance is insufficient, additional hand annotations would be required to improve the model's accuracy.

The integration of visual and auditory data in the ChildLens dataset enables a more detailed and comprehensive understanding of children's daily experiences. Complex activities such as pretend play and reading a book, which require both audio and video for accurate detection, exemplify the strength of this multimodal approach. While previous studies, such as those analyzing disfluencies in children's speech during computer game play (Yildirim & Narayanan, 2009), have demonstrated that combining visual and auditory information can improve performance, few studies have explored this in the context of children's naturalistic activities. With ChildLens, the combination of naturalistic data and multimodal analysis creates new opportunities for in-depth insights into children's cognitive, emotional, and social development, particularly for activities best captured through both modalities.

Despite its strengths, the ChildLens dataset also has its limitations. First, there is class imbalance, especially in underrepresented activity classes, which could affect model training and evaluation. More frequent activities, such as "playing with object" (1849 minutes) and "reading a book" (447 minutes), dominate the dataset, whereas less common activities like "dancing" (9 minutes) and "making music" (2 minutes) are scarcely represented. Similarly, activities like "playing without object" (48 minutes) and "watching something" (28 minutes) appear less frequently. This imbalance may lead to skewed model performance, making it harder to accurately classify rare activities. Possible solutions to this challenge could involve merging rare activity classes into broader categories or excluding them from model training, though these approaches may reduce the dataset's diversity. Other methods, such as resampling or augmentation, could help balance the dataset and improve model performance (Alani, Cosma, & Taherkhani, 2020; Spelmen & Porkodi, 2018). Second, there is sampling bias. Since the recordings are largely influenced by parental decisions about when and how often activities are captured, certain activities or settings may be overrepresented or underrepresented based on these preferences. Furthermore, the dataset primarily focuses on families from a mid-sized German city, limiting its geographic and cultural diversity. Expanding the dataset to include a broader range of families from different regions and cultures would enhance its generalizability and applicability to various research contexts.

The study of children's everyday experiences is crucial for understanding their cognitive, emotional, and social development. These daily interactions provide important insights into how children learn, grow, and engage with their environment. The ChildLens dataset makes a valuable contribution to this field by offering a rich multi-modal resource that captures children's experiences in naturalistic settings. With its comprehensive annotations and potential to automate the analysis of children's activities, the dataset enables researchers to develop, finetune and apply automized processing algorithms that help to scale-up the study of child development. By virtue of being an openly accessible resource, the ChildLens dataset creates new opportunities for understanding the complexities of early childhood development and provides a foundation for future research in this area.

Declarations

Funding

This work is funded by the Max Planck Society for the Advancement of Science and is supported by the German Research Foundation (DFG) project NFDI4DS under Grant No.: 460234259. We thank the participating families and children, and acknowledge the support from our research institutions and funding bodies.

Competing Interests

The authors have no competing interests to declare that are relevant to the content of this article.

Ethics approval

The study involving humans obtained approval from the Ethics Council of the Max Planck Society, Munich, Germany, falling under a packaged ethics application, "Uniquely human cultural diversity and its universal cognitive mechanisms" (Appl. No. 2021_45), and was approved by an internal ethics committee at the Max Planck Institute for Evolutionary Anthropology.

Code Availability

The code used to generate the benchmark performances is available on our GitHub page.

Authors' Contribution

- **Nele Pauline Suffo:** Conceptualization; Writing - Original Draft; Writing – Review & Editing; Data Curation; Formal analysis; Software

- **Pierre-Etienne Martin:** Conceptualization; Data Curation; Writing – Review & Editing
- **Anas Suffo:** Methodology; Software; Writing – Review & Editing;
- **Daniel Haun:** Conceptualization; Funding acquisition; Writing - Review & Editing;
- **Manuel Bohn:** Supervision; Conceptualization; Funding acquisition; Writing - Review & Editing

Open Practices Statement

The data and materials used in this study are openly available. The ChildLens dataset will be freely available for research purposes via an institutional repository (details listed on the ChildLens website). The code used to generate the benchmark performances is available on our GitHub page. This study was not preregistered.

References

Alani, A. A., Cosma, G., & Taherkhani, A. (2020). Classifying Imbalanced Multi-modal Sensor Data for Human Activity Recognition in a Smart Home using Deep Learning. *2020 International Joint Conference on Neural Networks (IJCNN)*, 1–8. Glasgow, United Kingdom: IEEE. https://doi.org/10.1109/IJCNN48605.2020.9207697

Baevski, A., Zhou, H., Mohamed, A., & Auli, M. (2020). Wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations. https://doi.org/10.48550/ARXIV.2006.11477

Bambach, S., Lee, S., Crandall, D. J., & Yu, C. (2015). Lending A Hand: Detecting Hands and Recognizing Activities in Complex Egocentric Interactions. *2015 IEEE International Conference on Computer Vision (ICCV)*, 1949–1957. Santiago, Chile: IEEE. https://doi.org/10.1109/ICCV.2015.226

Bang, J. Y., Kachergis, G., Weisleder, A., & Marchman, V. (2023). An automated classifier for periods of sleep and target-child-directed speech from LENA recordings. *Language Development Research*, *3.0*(1.0). https://doi.org/10.34842/XMRQ-ER43

Bergelson, E., Soderstrom, M., Schwarz, I.-C., Rowland, C. F., Ramírez-Esparza, N., R. Hamrick, L., … Cristia, A. (2023). Everyday language input and production in 1,001 children from six continents. *Proceedings of the National Academy of Sciences*, *120*(52), e2300671120. https://doi.org/10.1073/pnas.2300671120

Bredin, H. (2017). Pyannote.metrics: A Toolkit for Reproducible Evaluation, Diagnostic, and Error Analysis of Speaker Diarization Systems. *Interspeech 2017*, 3587–3591. ISCA: ISCA. https://doi.org/10.21437/interspeech.2017-411

Bredin, H. (2023). Pyannote.audio 2.1 speaker diarization pipeline: Principle, benchmark, and recipe. *INTERSPEECH 2023*, 1983–1987. ISCA. https://doi.org/10.21437/Interspeech.2023-105

Cao, Z., Hidalgo, G., Simon, T., Wei, S.-E., & Sheikh, Y. (2018). OpenPose: Realtime Multi-Person 2D Pose Estimation using Part Affinity Fields.

https://doi.org/10.48550/ARXIV.1812.08008

Carpendale, J., & Lewis, C. (2020). *What Makes Us Human: How Minds Develop through Social Interactions* (1st ed.). Routledge. https://doi.org/10.4324/9781003125105

Carreira, J., & Zisserman, A. (2017). Quo Vadis, Action Recognition? A New Model and the Kinetics Dataset. https://doi.org/10.48550/ARXIV.1705.07750

Chen, H., Alghowinem, S., Jang, S. J., Breazeal, C., & Park, H. W. (2023). Dyadic Affect in Parent-Child Multimodal Interaction: Introducing the DAMI-P2C Dataset and its Preliminary Analysis. *IEEE Transactions on Affective Computing*, *14*(4), 3345–3361. https://doi.org/10.1109/TAFFC.2022.3178689

Contributors, M. (2020). *OpenMMLab's Next Generation Video Understanding Toolbox and Benchmark*. Retrieved from urlhttps://github.com/open-mmlab/mmaction2

De Barbaro, K., & Fausey, C. M. (2022). Ten Lessons About Infants' Everyday Experiences. *Current Directions in Psychological Science*, *31*(1), 28–33. https://doi.org/10.1177/09637214211059536

Donnelly, S., & Kidd, E. (2021). The Longitudinal Relationship Between Conversational Turn-Taking and Vocabulary Growth in Early Language Development. *Child Development*, *92*(2), 609–625. https://doi.org/10.1111/cdev.13511

Ferjan Ramírez, N., Lytle, S. R., & Kuhl, P. K. (2020). Parent coaching increases conversational turns and advances infant language development. *Proceedings of the National Academy of Sciences*, *117*(7), 3484–3491. https://doi.org/10.1073/pnas.1921653117

Fraile, M., Calvo-Barajas, N., Apeiron, A. S., Varni, G., Lindblad, J., Sladoje, N., & Castellano, G. (2024). UpStory: The Uppsala Storytelling dataset. https://doi.org/10.48550/ARXIV.2407.04352

Ginsburg, K. R., Committee on Communications, & Committee on Psychosocial Aspects of Child and Family Health. (2007). The Importance of Play in Promoting Healthy Child Development and Maintaining Strong Parent-Child Bonds. *Pediatrics*, *119*(1), 182–191.

https://doi.org/10.1542/peds.2006-2697

Heilbron, F. C., Escorcia, V., Ghanem, B., & Niebles, J. C. (2015). ActivityNet: A large-scale video benchmark for human activity understanding. *2015 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 961–970. Boston, MA, USA: IEEE. https://doi.org/10.1109/CVPR.2015.7298698

Heyes, C. M. (2018). *Cognitive gadgets: The cultural evolution of thinking.* Cambridge (Mass.): Harvard University press.

Hofferth, S. L., & Sandberg, J. F. (2001). How American Children Spend Their Time. *Journal of Marriage and Family*, *63*(2), 295–308. https://doi.org/10.1111/j.1741-3737.2001.00295.x

Kazakos, E., Huh, J., Nagrani, A., Zisserman, A., & Damen, D. (2021). With a Little Help from my Temporal Context: Multimodal Egocentric Action Recognition. https://doi.org/10.48550/ARXIV.2111.01024

Lavechin, M., Bousbib, R., Bredin, H., Dupoux, E., & Cristia, A. (2020). An open-source voice type classifier for child-centered daylong recordings. https://doi.org/10.48550/ARXIV.2005.12656

Lin, T., Liu, X., Li, X., Ding, E., & Wen, S. (2019). BMN: Boundary-Matching Network for Temporal Action Proposal Generation. https://doi.org/10.48550/ARXIV.1907.09702

Long, B., Xiang, V., Stojanov, S., Sparks, R. Z., Yin, Z., Keene, G. E., … Frank, M. C. (2024, June 14). The BabyView dataset: High-resolution egocentric videos of infants' and young children's everyday experiences. https://doi.org/10.48550/arXiv.2406.10447

Piaget, J. (1964). Part I: Cognitive development in children: Piaget development and learning. *Journal of Research in Science Teaching*, *2*(3), 176–186. https://doi.org/10.1002/tea.3660020306

Plaquet, A., & Bredin, H. (2023). Powerset multi-class cross entropy loss for neural speaker diarization. *INTERSPEECH 2023*, 3222–3226. ISCA. https://doi.org/10.21437/Interspeech.2023-205

Redmon, J., Divvala, S., Girshick, R., & Farhadi, A. (2015). You Only Look Once: Unified, Real-Time Object Detection. https://doi.org/10.48550/ARXIV.1506.02640

Rehg, J. M., Abowd, G. D., Rozga, A., Romero, M., Clements, M. A., Sclaroff, S., … Ye, Z. (2013). Decoding Children's Social Behavior. *2013 IEEE Conference on Computer Vision and Pattern Recognition*, 3414–3421. Portland, OR, USA: IEEE. https://doi.org/10.1109/CVPR.2013.438

Rogoff, B., Dahl, A., & Callanan, M. (2018). The importance of understanding children's lived experience. *Developmental Review*, *50*, 5–15. https://doi.org/10.1016/j.dr.2018.05.006

Roy, B. C., Frank, M. C., DeCamp, P., Miller, M., & Roy, D. (2015). Predicting the birth of a spoken word. *Proceedings of the National Academy of Sciences*, *112*(41), 12663–12668. https://doi.org/10.1073/pnas.1419773112

Ruffman, T., Chen, L., Lorimer, B., Vanier, S., Edgar, K., Scarf, D., & Taumoepeau, M. (2023). Exposure to behavioral regularities in everyday life predicts infants' acquisition of mental state vocabulary. *Developmental Science*, *26*(4), e13343. https://doi.org/10.1111/desc.13343

Russakovsky, O., Deng, J., Su, H., Krause, J., Satheesh, S., Ma, S., … Fei-Fei, L. (2014). ImageNet Large Scale Visual Recognition Challenge. https://doi.org/10.48550/ARXIV.1409.0575

Smith, L. B., Jayaraman, S., Clerkin, E., & Yu, C. (2018). The Developing Infant Creates a Curriculum for Statistical Learning. *Trends in Cognitive Sciences*, *22*(4), 325–336. https://doi.org/10.1016/j.tics.2018.02.004

Spelmen, V. S., & Porkodi, R. (2018). A Review on Handling Imbalanced Data. *2018 International Conference on Current Trends Towards Converging Technologies (ICCTCT)*, 1–11. Coimbatore: IEEE. https://doi.org/10.1109/ICCTCT.2018.8551020

Sullivan, J., Mei, M., Perfors, A., Wojcik, E., & Frank, M. C. (2021). SAYCam: A Large, Longitudinal Audiovisual Dataset Recorded From the Infant's Perspective. *Open Mind*, *5*, 20–29. https://doi.org/10.1162/opmi_a_00039

Tomasello, M. (2009). *Cultural Origins of Human Cognition*. Cambridge: Harvard University Press.

Vygotsky, L. S. (1978). *Mind in Society: The Development of Higher Psychological Processes*. Cambridge, MA: Harvard University Press.

Xin, Q. (2020). *MN-Boundary-Matching-Network* [GitHub repository]. Retrieved from https://github.com/JJBOY/BMN-Boundary-Matching-Network

Yildirim, S., & Narayanan, S. (2009). Automatic Detection of Disfluency Boundaries in Spontaneous Speech of Children Using Audio–Visual Information. *IEEE Transactions on Audio, Speech, and Language Processing*, *17*(1), 2–12. https://doi.org/10.1109/TASL.2008.2006728

Zahra, A., Martin, P.-E., Bohn, M., & Haun, D. (2024). Computer Vision for Analyzing Children's Lived Experiences. In K. Arai (Ed.), *Intelligent Systems and Applications* (Vol. 823, pp. 376–383). Cham: Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-47724-9_25

Appendix

Activity Class and Location Statistics

Table 4

Number of video instances and the total duration in minutes for the number of currently annotated videos (n=192).

Category	Activity Class	Instance Count	Total Duration (min)
Audio	Child talking	12093	839.04
	Other person talking	10223	771.25
	Overheard Speech	3285	424.88
	Listening to music/audiobook	96	322.88
	Singing/Humming	361	104.01
Video	Drawing	83	366.30
	Crafting things	28	107
	Watching something	5	27.80
	Dancing	21	8.75
Multimodal	Playing with object	498	1849.30
	Reading a book	104	447.10
	Pretend play	68	182.26
	Playing without object	38	47.94
	Making music	3	2.47
Location	Livingroom	112	1590.99
	Playroom	71	998.73
	Other Room	49	559.88
	Hallway	18	42.47
	Bathroom	2	2.25

Benchmark Model Architectures

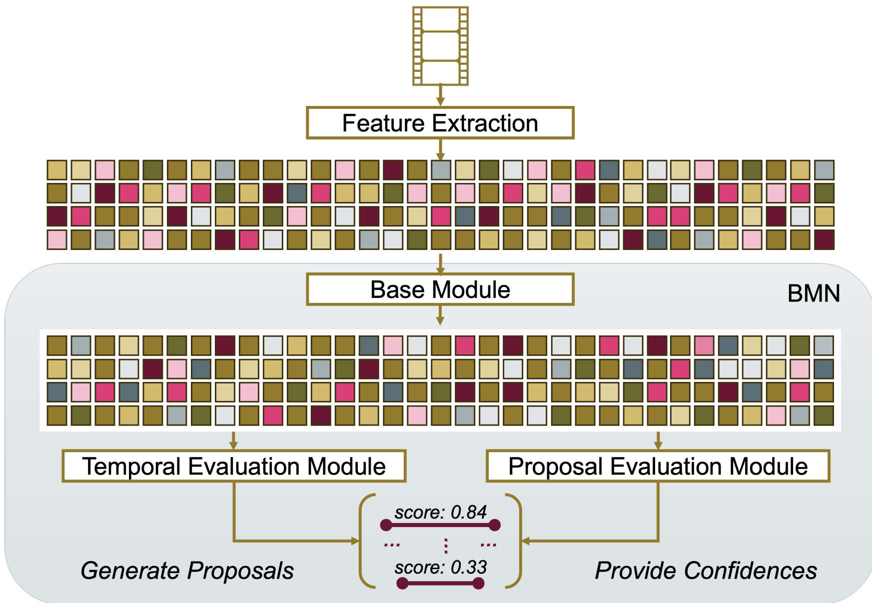

Figure 4. Boundary Matching Network Architecture (Lin, Liu, Li, Ding, & Wen, 2019)

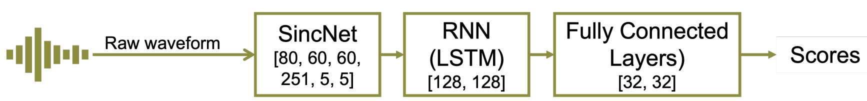

Figure 5. Voice Type Classifier Architecture (Lavechin, Bousbib, Bredin, Dupoux, & Cristia, 2020)